

HTML

Professional Complete Guide

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Chapter 1: Foundations

DOCTYPE, document structure, meta tags

Complete HTML5 Document Template

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
<head>
<!-- Character encoding – MUST be first -->
<meta charset="UTF-8">

<!-- Responsive viewport -->
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- IE compatibility (legacy) -->
<meta http-equiv="X-UA-Compatible" content="IE=edge">

<!-- SEO basics -->
<meta name="description" content="Page description (150-160 chars)">
<meta name="keywords" content="html, tutorial, web">
<meta name="author" content="Alice Smith">
<meta name="robots" content="index, follow">

<!-- Open Graph (social sharing) -->
<meta property="og:title" content="Page Title">
<meta property="og:description" content="Description">
<meta property="og:image" content="https://site.com/img.jpg">
<meta property="og:url" content="https://site.com/page">
<meta property="og:type" content="website">

<!-- Twitter Card -->
<meta name="twitter:card" content="summary_large_image">
<meta name="twitter:title" content="Page Title">

<!-- Favicon -->
<link rel="icon" href="/favicon.ico">
<link rel="icon" href="/icon.svg" type="image/svg+xml">
<link rel="apple-touch-icon" href="/apple-touch-icon.png">
<link rel="manifest" href="/site.webmanifest">

<!-- CSS -->
<link rel="stylesheet" href="styles.css">
<link rel="preconnect" href="https://fonts.googleapis.com">

<title>Page Title – Site Name</title>
</head>
<body>
<!-- Content here -->
<script src="app.js" defer></script>
</body>
</html>
```

Chapter 2: Text & Links

Headings, paragraphs, formatting, hyperlinks

Text Elements

```
<!-- Headings: h1 should be unique per page for SEO -->
<h1>Main Page Title</h1>
<h2>Section Heading</h2>
<h3>Subsection</h3> <h4>Sub-sub</h4> <h5>...</h5> <h6>...</h6>

<!-- Paragraph -->
<p>A paragraph. HTML collapses whitespace. Extra spaces ignored.</p>

<!-- Inline text semantics -->
<strong>Important bold</strong> <b>Stylistic bold</b>
<em>Emphasized italic</em> <i>Idiomatic italic</i>
<u>Underlined</u> <s>Strikethrough</s>
<mark>Highlighted text</mark> <small>Fine print</small>
<del>Deleted</del> <ins>Inserted</ins>
<sup>superscript</sup> <sub>subscript</sub>
<abbr title="HyperText Markup Language">HTML</abbr>
<cite>Book Title</cite>
<q>Short inline quote</q>
<code>inline code</code>
<kbd>Ctrl+C</kbd> <!-- keyboard input -->
<samp>output text</samp> <!-- program output -->
<var>x</var> <!-- math variable -->
<time datetime="2026-05-10">May 10, 2026</time>

<!-- Block quotes -->
<blockquote cite="https://source.com">
<p>"The only way to do great work is to love what you do."</p>
<footer>– <cite>Steve Jobs</cite></footer>
</blockquote>

<!-- Preformatted text -->
<pre><code>
function hello() {
  console.log("Hello");
}
</code></pre>

<br> <!-- line break – use sparingly -->
<hr> <!-- thematic break / horizontal rule -->
<wbr> <!-- word break opportunity -->
```

Hyperlinks

```
<!-- External link -->
<a href="https://example.com">Visit Example</a>

<!-- Open in new tab (always add rel for security) -->
<a href="https://example.com" target="_blank"
rel="noopener noreferrer">Open in new tab</a>

<!-- Internal page link -->
<a href="/about">About Us</a>
<a href="./page.html">Relative link</a>

<!-- Anchor / scroll to section -->
<a href="#contact">Jump to Contact</a>
<section id="contact"><h2>Contact</h2></section>

<!-- Email / phone / SMS -->
<a href="mailto:info@example.com">Email us</a>
<a href="mailto:info@example.com?subject=Hello&body=Hi there">
<a href="tel:+491234567890">Call us</a>
<a href="sms:+491234567890">Text us</a>

<!-- Download link -->
<a href="/report.pdf" download>Download PDF</a>
<a href="/report.pdf" download="annual-report-2026.pdf">Download</a>

<!-- Image as link -->
<a href="/gallery"></a>
```

Chapter 3: Lists & Tables

All list types and complete table markup

Lists

```
<!-- Unordered list -->
<ul>
<li>Item one</li>
<li>Item two
<ul> <!-- nested list -->
<li>Sub-item</li>
</ul>
</li>
</ul>

<!-- Ordered list -->
<ol type="1"> <!-- 1, A, a, I, i -->
<li>First step</li>
<li value="5">This is step 5</li> <!-- custom start -->
</ol>
<ol start="3" reversed> <!-- start at 3, count down -->
<li>Step 3</li> <li>Step 2</li>
</ol>

<!-- Description list -->
<dl>
<dt>HTML</dt>
<dd>HyperText Markup Language – the structure of web pages.</dd>
<dt>CSS</dt>
<dd>Cascading Style Sheets – the presentation layer.</dd>
</dl>
```

Tables — Full Reference

```
<table>
<caption>Monthly Sales Data 2026</caption>
<colgroup>
<col span="1" style="background:#f0f0f0">
<col span="3">
</colgroup>
<thead>
<tr>
<th scope="col">Month</th>
<th scope="col">Product A</th>
<th scope="col">Product B</th>
<th scope="col">Total</th>
</tr>
</thead>
<tbody>
<tr>
<th scope="row">January</th>
<td>1200</td>
<td>850</td>
<td>2050</td>
</tr>
<tr>
<td colspan="3">Q1 Subtotal</td> <!-- span columns -->
<td>6150</td>
</tr>
<tr>
<td rowspan="2">Q2</td> <!-- span rows -->
<td>Apr: 1400</td>
<td>Apr: 900</td>
<td>2300</td>
</tr>
</tbody>
<tfoot>
<tr>
<th>Total</th>
<td>14400</td>
<td>9800</td>
<td>24200</td>
</tr>
</tfoot>
</table>
```

Chapter 4: Images & Media

img, picture, audio, video, iframe, SVG

Images

```
<!-- Basic image -->


<!-- Responsive image with srcset -->


<!-- Art direction with picture -->
<picture>
<source media="(min-width: 1200px)" srcset="hero-large.jpg">
<source media="(min-width: 600px)" srcset="hero-medium.jpg">
<source type="image/webp" srcset="hero.webp">
 <!-- fallback -->
</picture>

<!-- Figure with caption -->
<figure>

<figcaption>Figure 1: Q1 2026 Sales by Region</figcaption>
</figure>
```

Audio & Video

```
<!-- Audio player -->
<audio controls preload="metadata">
<source src="podcast.mp3" type="audio/mpeg">
<source src="podcast.ogg" type="audio/ogg">
<p>Your browser does not support HTML5 audio.
<a href="podcast.mp3">Download instead</a>.</p>
</audio>

<!-- Video player -->
<video width="854" height="480"
controls autoplay muted loop poster="thumbnail.jpg"
preload="auto">
<source src="video.mp4" type="video/mp4">
<source src="video.webm" type="video/webm">
<track src="subtitles_en.vtt" kind="subtitles"
srclang="en" label="English" default>
<track src="subtitles_de.vtt" kind="subtitles"
srclang="de" label="Deutsch">
Your browser does not support HTML5 video.
</video>

<!-- YouTube / external embed -->
<iframe width="560" height="315"
src="https://www.youtube.com/embed/VIDEO_ID?rel=0&modestbranding=1"
title="Video description"
frameborder="0"
allow="accelerometer; autoplay; clipboard-write; encrypted-media"
allowfullscreen
loading="lazy">
</iframe>
```


Chapter 5: Forms — Complete

Every input type, validation, and best practices

All Input Types

Form — Part 1: Input types

```
<form action="/register" method="POST" novalidate
  enctype="multipart/form-data" autocomplete="on">

<!-- Text inputs -->
<label for="fn">First Name</label>
<input type="text" id="fn" name="fname" required minlength="2">
<input type="email" name="email" required autocomplete="email">
<input type="password" name="pass" required minlength="8">
<input type="tel" name="phone" pattern="[+][0-9]{10,15}">
<input type="url" name="website" placeholder="https://">
<input type="search" name="q" placeholder="Search...">

<!-- Numeric & Range -->
<input type="number" name="age" min="1" max="120" step="1">
<input type="range" name="score" min="0" max="100" step="5">

<!-- Date & Time -->
<input type="date" name="dob">
<input type="time" name="appt">
<input type="datetime-local" name="event">
<input type="month" name="month">
<input type="color" name="theme" value="#264DE4">
<input type="file" name="avatar" accept="image/*" multiple>
<input type="hidden" name="csrf_token" value="abc123">
```

Form — Part 2: Select, textarea, checkboxes

```
<!-- Select with optgroups -->
<select name="country" required>
<option value="">Choose...</option>
<optgroup label="Europe">
<option value="de">Germany</option>
<option value="fr">France</option>
</optgroup>
</select>

<!-- Datalist autocomplete -->
<input list="langs" name="language">
<datalist id="langs">
<option value="JavaScript">
<option value="Python">
<option value="PHP">
</datalist>

<!-- Textarea -->
<textarea name="bio" rows="4" maxlength="500"
placeholder="Tell us about you..."></textarea>

<!-- Checkbox & Radio -->
<input type="checkbox" id="terms" name="agree" value="yes" required>
<label for="terms">I agree to the Terms</label>

<fieldset>
<legend>Gender</legend>
<label><input type="radio" name="gender" value="m"> Male</label>
<label><input type="radio" name="gender" value="f"> Female</label>
</fieldset>

<!-- Buttons -->
<button type="submit">Register</button>
<button type="reset">Clear</button>
<button type="button" onclick="preview()">Preview</button>
</form>
```

Chapter 6: Semantic HTML5

Meaningful elements for accessibility and SEO

Semantic Layout Structure

```
<body>
<header>
<a href="/" class="logo"></a>
<nav aria-label="Main navigation">
<ul>
<li><a href="/" aria-current="page">Home</a></li>
<li><a href="/about">About</a></li>
<li>
<a href="/services" aria-haspopup="true">Services</a>
<ul> <!-- dropdown -->
<li><a href="/web-design">Web Design</a></li>
</ul>
</li>
</ul>
</nav>
</header>

<main id="main-content">
<article>
<header>
<h1>Article Title</h1>
<p>By <address><a href="mailto:a@b.com">Alice</a></address>
on <time datetime="2026-05-10">May 10, 2026</time></p>
</header>

<section aria-label="Introduction">
<h2>Introduction</h2>
<p>First paragraph...</p>
</section>

<aside aria-label="Related info">
<h3>Did you know?</h3>
<p>Fun fact...</p>
</aside>

<footer>
<p>Tags: <a href="/tag/html">HTML</a></p>
</footer>
</article>

<aside aria-label="Sidebar">
<section><h2>Recent Posts</h2></section>
</aside>
</main>

<footer>
<nav aria-label="Footer navigation">
<a href="/privacy">Privacy</a> |
<a href="/contact">Contact</a>
</nav>
<p><small>&copy; 2026 Company Name. All rights reserved.</small></p>
</footer>
</body>
```

Element	Purpose	Use When
<header>	Introductory content	Top of page or section
<nav>	Navigation links	Site/page navigation menus
<main>	Primary content (once per page)	Central content area

<article>	Self-contained content	Blog post, news item, comment
<section>	Thematic grouping	Chapter, topic group
<aside>	Tangentially related	Sidebar, callout, ads
<footer>	Footer content	Copyright, links, address
<figure>	Self-contained media	Images, diagrams, code
<figcaption>	Caption for figure	Below/above figure element
<details>	Disclosure widget	FAQ, expandable info
<summary>	Summary for details	Visible toggle text
<dialog>	Native dialog/modal	Pop-ups, confirmations

Chapter 7: Canvas & SVG

Drawing graphics with HTML5

HTML5 Canvas

```
<canvas id="myCanvas" width="600" height="400"
aria-label="Drawing canvas"></canvas>

<script>
const canvas = document.getElementById("myCanvas");
const ctx = canvas.getContext("2d");

// Rectangle
ctx.fillStyle = "#264DE4";
ctx.fillRect(10, 10, 200, 100); // filled
ctx.strokeStyle = "red"; ctx.lineWidth = 3;
ctx.strokeRect(220, 10, 200, 100); // outlined
ctx.clearRect(50, 50, 50, 50); // clear area

// Text
ctx.font = "bold 24px Arial";
ctx.fillStyle = "white";
ctx.textAlign = "center";
ctx.fillText("Hello Canvas!", 300, 200);
ctx.strokeText("Outlined", 300, 240);

// Path / Lines
ctx.beginPath();
ctx.moveTo(50, 300); ctx.lineTo(200, 350); ctx.lineTo(350, 280);
ctx.closePath();
ctx.stroke();

// Circle / Arc
ctx.beginPath();
ctx.arc(450, 200, 80, 0, Math.PI * 2); // full circle
ctx.fillStyle = "gold"; ctx.fill();

// Gradient
const grad = ctx.createLinearGradient(0, 0, 600, 0);
grad.addColorStop(0, "ff0000");
grad.addColorStop(1, "0000ff");
ctx.fillStyle = grad;
ctx.fillRect(0, 350, 600, 50);

// Image
const img = new Image();
img.onload = () => ctx.drawImage(img, 10, 10, 100, 100);
img.src = "photo.jpg";
</script>
```

Inline SVG

```
<!-- SVG directly in HTML -->
<svg width="400" height="300" xmlns="http://www.w3.org/2000/svg"
role="img" aria-label="Simple SVG diagram">
<title>SVG Shapes</title>

<!-- Shapes -->
<rect x="10" y="10" width="150" height="80" rx="10"
fill="#264DE4" stroke="navy" stroke-width="2"/>
<circle cx="250" cy="50" r="40" fill="gold" opacity="0.8"/>
<ellipse cx="350" cy="50" rx="50" ry="30" fill="tomato"/>
<line x1="10" y1="150" x2="390" y2="150"
stroke="#333" stroke-width="2" stroke-dasharray="8,4"/>

<!-- Polygon & Polyline -->
<polygon points="200,120 250,200 150,200" fill="lime"/>
<polyline points="10,250 100,200 200,240 300,180 390,230"
fill="none" stroke="purple" stroke-width="3"/>

<!-- Text -->
<text x="50" y="280" font-size="20" font-family="Arial"
fill="#333" font-weight="bold">SVG Text</text>

<!-- Path (complex shapes) -->
<path d="M 300,250 C 320,200 380,200 400,250 S 460,300 480,250"
fill="none" stroke="red" stroke-width="3"/>
</svg>
```

Chapter 8: Web Storage & APIs

localStorage, sessionStorage, Geolocation

Web Storage API

```
// localStorage – persists across sessions
localStorage.setItem("theme", "dark");
localStorage.setItem("user", JSON.stringify({name:"Alice",id:42}));

const theme = localStorage.getItem("theme"); // "dark"
const user = JSON.parse(localStorage.getItem("user"));

localStorage.removeItem("theme");
localStorage.clear(); // remove ALL items
localStorage.length; // number of items

// sessionStorage – cleared when tab closes
sessionStorage.setItem("step", "2");
sessionStorage.getItem("step"); // "2"

// Storage event (fires in OTHER tabs/windows)
window.addEventListener("storage", (e) => {
  console.log("Key:", e.key, "New:", e.newValue);
});
```

Geolocation API

```
// Get current position
if ("geolocation" in navigator) {
  navigator.geolocation.getCurrentPosition(
    (position) => {
      const { latitude, longitude, accuracy } = position.coords;
      console.log(`Lat: ${latitude}, Lng: ${longitude}`);
    },
    (error) => {
      switch(error.code) {
        case error.PERMISSION_DENIED:
          console.log("Permission denied"); break;
        case error.POSITION_UNAVAILABLE:
          console.log("Position unavailable"); break;
      }
    },
    { enableHighAccuracy: true, timeout: 5000, maximumAge: 60000 }
  );
}

// Watch position (continuous)
const watchId = navigator.geolocation.watchPosition(successFn);
navigator.geolocation.clearWatch(watchId); // stop watching
} else {
  console.log("Geolocation not supported");
}
```


Chapter 9: Accessibility (ARIA)

Making HTML accessible for everyone

ARIA Roles & Attributes

ARIA (Accessible Rich Internet Applications) adds semantic meaning to HTML elements for screen readers and assistive technology.

```
<!-- ARIA Roles -->
<div role="button" tabindex="0" onclick="..." onkeypress="...">
Click me
</div>

<!-- Better: use native elements when possible -->
<button onclick="...">Click me</button>

<!-- Labels -->
<input type="text" aria-label="Search the site">
<input type="text" aria-labelledby="search-label">
<span id="search-label">Search</span>

<!-- Descriptions -->
<input type="password" aria-describedby="pw-hint">
<p id="pw-hint">Must be 8+ characters with a number.</p>

<!-- States -->
<button aria-pressed="false">Toggle</button>
<button aria-expanded="false" aria-controls="menu">Menu</button>
<div id="menu" aria-hidden="true">...</div>

<!-- Live regions (dynamic content announcements) -->
<div aria-live="polite" aria-atomic="true">
<p id="status-message"></p> <!-- updated dynamically -->
</div>

<!-- Landmark roles (use semantic HTML instead where possible) -->
<div role="banner"> <!-- use <header> -->
<div role="navigation"> <!-- use <nav> -->
<div role="main"> <!-- use <main> -->
<div role="contentinfo"> <!-- use <footer> -->

<!-- Skip link (keyboard accessibility) -->
<a href="#main-content" class="skip-link">Skip to main content</a>
```

ARIA Attribute	Purpose	Example
aria-label	Names an element	aria-label='Close dialog'
aria-labelledby	References labelling element	aria-labelledby='title-id'
aria-describedby	References description	aria-describedby='hint-id'
aria-hidden	Hides from screen readers	aria-hidden='true'
aria-live	Announces dynamic changes	aria-live='polite'

aria-expanded	Discloses expand state	aria-expanded='false'
aria-controls	References controlled element	aria-controls='menu-id'
aria-pressed	Toggle button state	aria-pressed='true'
aria-required	Marks required field	aria-required='true'
aria-invalid	Marks invalid field	aria-invalid='true'
aria-disabled	Marks disabled state	aria-disabled='true'
role	Defines element role	role='dialog'

Chapter 10: Exercises

20 HTML exercises covering all topics

Exercises

FILL

Ex 1. What HTML tag defines the largest heading?

■ **Answer:** `<h1>` – Use only one per page for proper SEO structure.

FILL

Ex 2. What attribute makes a form field mandatory?

■ **Answer:** `required` – e.g. `<input type='text' required>`;

TRUEFALSE

Ex 3. `<b>` and `<strong>` have the same semantic meaning. (True/False)

■ **Answer:** FALSE – `<strong>` means important (semantic). `<b>` is stylistic bold only. Screen readers treat `<strong>` differently.

WRITE

Ex 4. Write the HTML for a responsive image that serves different sizes.

■ **Answer:** `<img src='img-800.jpg' srcset='img-400.jpg 400w, img-800.jpg 800w' sizes='(max-width:600px) 100vw, 800px' alt='description' loading='lazy'>`

FILL

Ex 5. What attribute should be added to an `iframe` embedding YouTube for accessibility?

■ **Answer:** `title` attribute – e.g. `title='How to code in Python – Tutorial'`

WRITE

Ex 6. Write a complete accessible form with email, password, and submit button.

■ **Answer:** `<form><label for='e'>Email</label><input type='email' id='e' name='email' required autocomplete='email'><label for='p'>Password</label><input type='password' id='p' name='pass' required minlength='8'><button type='submit'>Login</button></form>`

FILL

Ex 7. What is the purpose of the `<figure>` and `<figcaption>` elements?

■ **Answer:** `<figure>` groups self-contained content (image, diagram, code). `<figcaption>` provides a visible caption. Together they associate the caption semantically with the content.

EXPLAIN

Ex 8. What is the difference between localStorage and sessionStorage?

■ **Answer:** localStorage persists data until explicitly cleared – survives browser restart. sessionStorage is cleared when the tab/window closes. Both store string key-value pairs, max ~5-10MB each.

WRITE

Ex 9. Write HTML5 Canvas code to draw a red circle at position (200,200) with radius 60.

■ **Answer:**

```
const ctx=canvas.getContext('2d'); ctx.beginPath();  
ctx.arc(200,200,60,0,Math.PI*2); ctx.fillStyle='red'; ctx.fill();
```

FILL

Ex 10. What ARIA attribute hides an element from screen readers?

■ **Answer:** aria-hidden='true'

TRUE/FALSE

Ex 11. The <main> element can appear multiple times per page. (True/False)

■ **Answer:** FALSE – <main> should appear only once per page. It identifies the primary content area.

WRITE

Ex 12. Write the meta tags needed for a complete Open Graph social sharing setup.

■ **Answer:**

```
<meta property='og:title' content='Title'>  
<meta property='og:description' content='Desc'>  
<meta property='og:image' content='https://site.com/img.jpg'>  
<meta property='og:url' content='https://site.com/page'>  
<meta property='og:type' content='website'>
```

FILL

Ex 13. What input type should be used for a color picker?

■ **Answer:**

```
<input type='color' name='theme' value='#ff0000'>
```

EXPLAIN

Ex 14. Why should you use semantic HTML elements instead of generic divs?

■ **Answer:** Semantics improve: (1) Accessibility – screen readers understand structure, (2) SEO – search engines understand content hierarchy, (3) Maintainability – easier to read and maintain code, (4) Default browser styling and behavior.

WRITE

Ex 15. Write a table with 3 columns (Name, Score, Grade) and 3 rows of data, including thead and tfoot.

■ **Answer:** `<table><thead><tr><th>Name</th><th>Score</th><th>Grade</th></tr></thead><tbody><tr><td>Alice</td><td>95</td><td>A</td></tr><tr><td>Bob</td><td>82</td><td>B</td></tr><tr><td>Carol</td><td>74</td><td>C</td></tr></tbody><tfoot><tr><td>Average</td><td>83.7</td><td>B</td></tr></tfoot></table>`

FILL

Ex 16. What attribute on `<script>` delays execution until the HTML is fully parsed?

■ **Answer:** `defer` – e.g. `<script src='app.js' defer></script>`

WRITE

Ex 17. Write the HTML for a dropdown select with 3 options grouped into 2 optgroups.

■ **Answer:** `<select name='city'><optgroup label='Germany'><option value='ber'>Berlin</option><option value='mun'>Munich</option></optgroup><optgroup label='France'><option value='par'>Paris</option></optgroup></select>`

EXPLAIN

Ex 18. What is the purpose of the alt attribute on images?

■ **Answer:** `alt` provides alternative text for: (1) screen readers to describe the image to visually impaired users, (2) display when image fails to load, (3) SEO – search engines index alt text. Decorative images should have `alt=''` (empty).

TRUEFALSE

Ex 19. The `defer` and `async` attributes on `<script>` work the same way. (True/False)

■ **Answer:** FALSE – `defer`: loads in parallel, executes IN ORDER after HTML parsed. `async`: loads in parallel, executes IMMEDIATELY when loaded (may be out of order). Use `defer` for most scripts.

WRITE

Ex 20. Write an accessible navigation landmark with 4 links and a proper aria-label.

■ **Answer:** `<nav aria-label='Main navigation'><ul><li>a href='/' aria-current='page'>Home</a></li><li>a href='/about'>About</a></li><li>a href='/services'>Services</a></li><li>a href='/contact'>Contact</a></li></ul></nav>`